Phase 1 Project Submission

DSO 576 | Python and Agentic Coding for Business Analytics

Note: We switched our topic from Amazon reviews to a short-term rental marketplace for Los Angeles, and the dashboard focuses on helping users compare neighborhoods, prices, and availability for booking or listing decisions.

1 Establish the data fit

I opened the public Airbnb listings file for the Los Angeles market, stored in the workspace as **listings.csv**, from the Inside Airbnb public data archive: https://insideairbnb.com/get-the-data/.

Source and scope. The file contains one row per Airbnb listing. The sample has **43,932 rows** and **19 columns**, covering residential and short-stay listings across Los Angeles-area neighborhoods such as Hollywood, Venice, Santa Monica, Downtown, Long Beach, Pasadena, Beverly Hills, and Glendale. This is a cross-sectional snapshot, not a time-series dataset, so it supports market comparison rather than a dated trend analysis.

Column and source table	What it contains	How we will use it
neighbourhood in listings	Neighborhood name for each listing	Filter by area and compare local pricing and demand.
room_type in listings	Entire home/apartment, private room, hotel room, or shared room	Compare market segments and evaluate booking-type tradeoffs.
price in listings	Nightly price in USD	Main affordability metric and central filter for decision support.
availability_365 in listings	Number of nights available over the next 365 days	Proxy for occupancy and supply; spot high-demand areas.
number_of_reviews in listings	Total review count	Measure proof of market traction and listing activity.
reviews_per_month in listings	Average monthly review activity	Estimate recent demand and recency of guest interest.
latitude and longitude in listings	Geographic location for each listing	Show a map of density and pricing patterns by area.

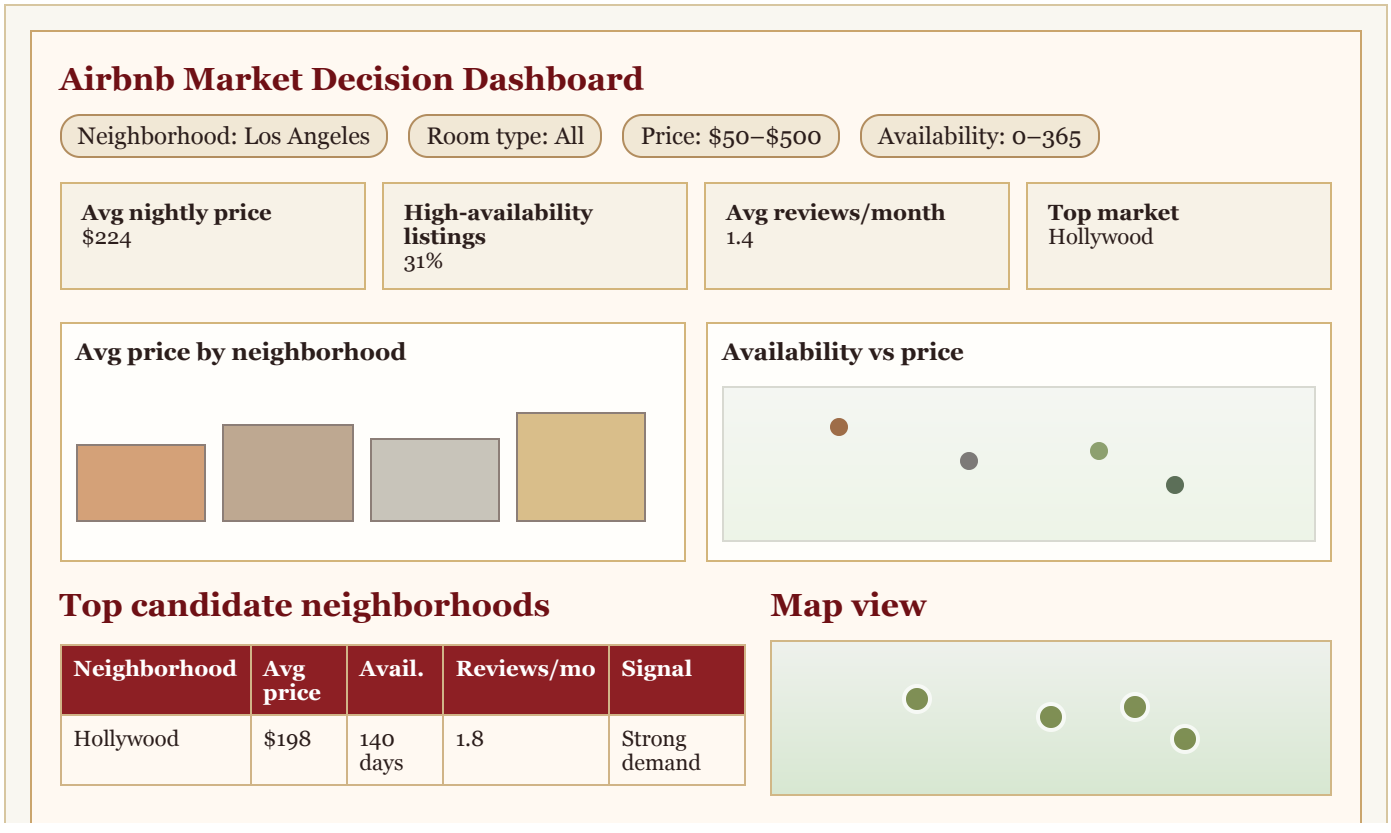
Column and source table	What it contains	How we will use it
minimum_nights in listings	Minimum stay requirement	Understand booking flexibility and host expectations.

What the data can support. The available data supports a decision dashboard for either travelers or hosts. It can answer questions such as which neighborhoods offer the best balance of price, availability, and review activity, which room types are most competitive, and where supply may be more constrained. The dashboard could support neighborhood and room-type filters, a price-vs-availability scatterplot, average price by area, a ranked listing table, and a map showing listing density and average nightly rate.

Important missing information includes seasonal demand, amenities, host response time, cancellation policy, and listing photos. Those gaps are manageable: a smaller initial scope can focus on price, neighborhood, room type, availability, and review volume. A second phase could add amenities and calendar data from a separate source to improve conversion and recommendation quality.

2 Sketch the dashboard

Below is the main screen concept: a user can choose a neighborhood, room type, and price range, then immediately compare value, demand, and availability before deciding where to stay or what type of listing to launch.



Neighborhood	Avg price	Avail.	Reviews/mo	Signal
Venice	$260	110 days	1.4	Premium market
Long Beach	$175	180 days	1.2	Value option

Decision help: “For a traveler with a $200 budget, Hollywood and Long Beach show the best value-to-demand tradeoff; for a host, Venice has higher revenue potential but also greater competition.”

Annotations: the sidebar filters rely on *neighbourhood*, *room_type*, and *price*; the charts use *price*, *availability_365*, and *reviews_per_month*; the map is built from *latitude* and *longitude*; and the decision signal is a calculated value score combining lower price, higher availability, and stronger review activity.

Repo link: https://github.com/cynthiaqgao/project